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Report

Partial Fulfilment of the Requirement for the Subject

MM711 Programming for Analytics with SAS

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Contents

1. Introduction

This report examines how different housing characteristics influence the **sale price** of property. The case study uses two datasets. Dataset 1 contains housing characteristics such as **living area**, **garage area**, **garage type**, number of bedrooms and bathrooms. Dataset 2 contains sale information such as the year the house was built, the year the house was sold and the price.

The goal of this analysis is to determine which property features best predict a house sale price. The dependent variable for this study will be **SoldPrice**. The independent variables are **LivingArea_sm**, **Garage_sm**, **Garage_Type**, **Bedroom**, **Bathroom**, **ConstructionYear**, and **YearSold**. The additional variable **HouseAge** was derived by subtracting ConstructionYear from YearSold.

Its main research question is:

What is the effect of housing characteristics on sale price?

In order to answer this question, the two datasets were merged, checked for data quality, cleaned, explored descriptively using statistics and graphs, and compared using **correlation** and **regression analysis**. This met the requirement of collecting and cleaning data, visualizing results, and performing a statistical analysis of the data.

2. Data Preparation and Cleaning

The analysis started by importing the two Excel files into SAS. The characteristics dataset contained **200 observations** and **6 variables**.

The CONTENTS Procedure

Data Set Name	WORK.CHARACTERISTICS	**Observations**	200
Member Type	DATA	**Variables**	6
Engine	V9	**Indexes**	0
Created	09/05/2026 11:22:53	**Observation Length**	48
Last Modified	09/05/2026 11:22:53	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

The house price dataset contained **210 observations** and **4 variables**.

The CONTENTS Procedure

Data Set Name	WORK.HOUSEPRICE	**Observations**	210
Member Type	DATA	**Variables**	4
Engine	V9	**Indexes**	0
Created	09/05/2026 11:22:53	**Observation Length**	32
Last Modified	09/05/2026 11:22:53	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			

The common variable in both datasets was **Reference**, which identifies each house. This variable was used to merge the two datasets.

Before merging, both datasets were sorted by Reference. Duplicate checks were then carried out. The characteristics dataset did not show duplicate records in the SAS output. The house price dataset did contain duplicate records. The duplicate references were **27260 to 27269**.

Duplicate References Removed from House Price Dataset

House Reference Number	Construction Year	Year Sold	Sale Price (£)
27260	1997	2007	129200
27261	1986	2007	136000
27262	1979	2007	153880
27263	1973	2007	90800
27264	2005	2007	161600
27265	2003	2007	132400
27266	1977	2007	96600
27267	1976	2007	174400
27268	1978	2007	171000
27269	1963	2007	101600

These appeared twice with the same sale information. The duplicate records were removed before the final analysis. After removing duplicate records, the two datasets were merged using Reference. Only houses that appeared in both datasets were kept. This produced a combined dataset with **200 records** before cleaning.

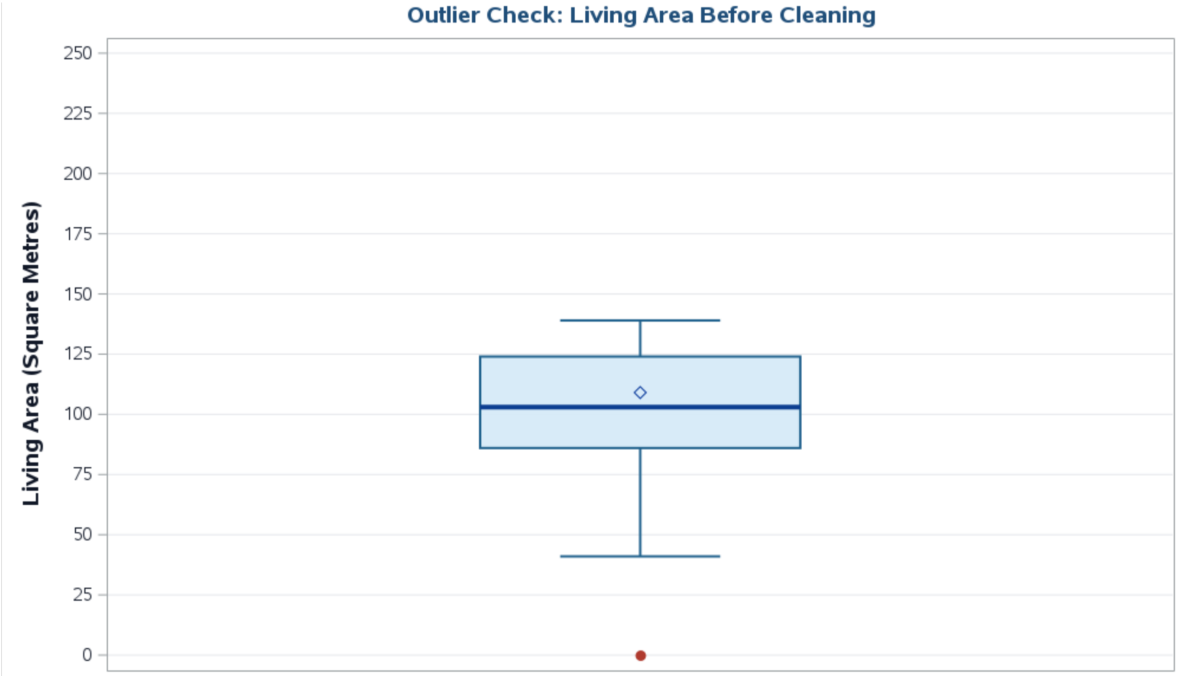
Several data quality issues were identified.

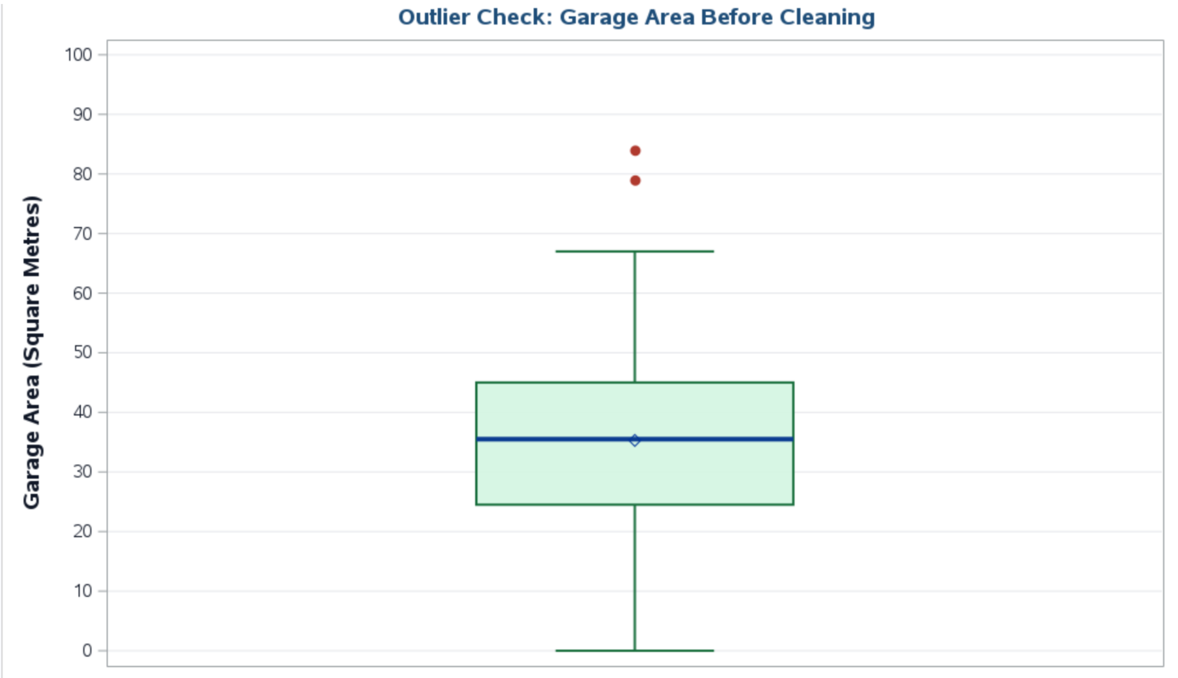
Missing Value Check Before Cleaning

The MEANS Procedure

Variable	Label	N	N Miss
ConstructionYear	Construction Year	200	0
YearSold	Year Sold	197	3
SoldPrice	Sale Price (£)	199	1
LivingArea_sm	Living Area (Square Metres)	200	0
Garage_sm	Garage Area (Square Metres)	200	0
Bedroom	Number of Bedrooms	200	0
Bathroom	Number of Bathrooms	200	0

1. YearSold had **3 missing values**.
2. SoldPrice had **1 missing value**. Since SoldPrice is the outcome variable, this record could not be used in the main analysis.
3. One record had **LivingArea_sm = 0**, which is not realistic for a house.
4. Reference **27183** had **LivingArea_sm = 1330** and **Garage_sm = 380**. These values were far larger than the rest of the dataset.





The boxplots before cleaning also showed these as **extreme outliers**. This record was removed because it appeared to be a **data entry error**.

Garage type also needed cleaning. The original **Garage_Type** variable included Attached, Detached, NA, and blank values. The brief states that **NA means information is not available**. In the cleaned dataset, blank garage types were recorded as **Unknown**. Records with **Garage_Type = NA** and **Garage_sm = 0** were recorded as **No garage/NA**. A second variable, **Garage_Status**, was also created to show whether the property had garage space recorded.

Garage Type Values Before Cleaning

The FREQ Procedure

Garage Type				
Garage_Type	**Frequency**	**Percent**	**Cumulative Frequency**	**Cumulative Percent**
	42	21.00	42	21.00
Attached	83	41.50	125	62.50
Detached	52	26.00	177	88.50
NA	23	11.50	200	100.00

After all cleaning steps, the final dataset contained **194 complete observations**. The SAS output showed **no missing values** after cleaning. This means the final analysis was carried out on a clean dataset with complete values for all variables used in the main analysis.

Missing Value Check After Cleaning

The MEANS Procedure

Variable	Label	N	N Miss
ConstructionYear	Construction Year	194	0
YearSold	Year Sold	194	0
SoldPrice	Sale Price (£)	194	0
LivingArea_sm	Living Area (Square Metres)	194	0
Garage_sm	Garage Area (Square Metres)	194	0
Bedroom	Number of Bedrooms	194	0
Bathroom	Number of Bathrooms	194	0
HouseAge	House Age at Sale	194	0

Final Cleaned Dataset Saved in MM711 Library

The CONTENTS Procedure

Data Set Name	MM711.HOUSING_CLEAN	**Observations**	194
Member Type	DATA	**Variables**	13
Engine	V9	**Indexes**	0
Created	09/05/2026 11:22:57	**Observation Length**	128
Last Modified	09/05/2026 11:22:57	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	1022
Obs in First Data Page	194
Number of Data Set Repairs	0
Filename	/home/u64455556/MM711/hpd18 MYDATA/housing_clean.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	2892053877
Access Permission	rw-r--r--
Owner Name	u64455556
File Size	256KB

Engine/Host Dependent Information	
File Size (bytes)	262144

Variables in Creation Order

#	Variable	Type	Len	Format	Informat	Label
1	Reference	Num	8	BEST.		House Reference Number
2	ConstructionYear	Num	8	BEST.		Construction Year
3	YearSold	Num	8	BEST.		Year Sold
4	SoldPrice	Num	8	POUND12.2		Sale Price (£)
5	LivingArea_sm	Num	8	BEST.		Living Area (Square Metres)
6	Garage_sm	Num	8	BEST.		Garage Area (Square Metres)
7	Garage_Type	Char	8	$8.	$8.	Garage Type
8	Bedroom	Num	8	BEST.		Number of Bedrooms
9	Bathroom	Num	8	BEST.		Number of Bathrooms
10	GarageType_Clean	Char	20			Garage Type
11	Garage_Status	Char	15			Garage Status
12	HouseAge	Num	8			House Age at Sale
13	LogSoldPrice	Num	8			Log of Sale Price

The final cleaned dataset was also saved in the **MM711 library** for later analysis.

3. Descriptive Statistics

The cleaned dataset showed a wide range in house prices. The mean sale price was **£106,155.08**. The median sale price was **£104,800**. The lowest sale price was **£28,000** and the highest sale price was **£176,000**. This shows that the dataset included both low-priced and high-priced properties.

The smallest living area in the cleaned dataset was **41 m^2**. The largest living area was **138 m^2**, while the average living area was **103.03 m^2**. The mean garage area was **33.83 m^2**, with values ranging from **0** to **84 m^2**.

Most houses had either **2** or **3 bedrooms**. The mean number of bedrooms was **2.51**. The bedroom frequency table showed that **16 houses** had 1 bedroom, **68 houses** had 2 bedrooms, **106 houses** had 3 bedrooms, and **4 houses** had 4 bedrooms.

Frequency Table: Number of Bedrooms

The FREQ Procedure

Number of Bedrooms				
Bedroom	Frequency	Percent	Cumulative Frequency	Cumulative Percent

[redacted]

Number of Bedrooms				
Bedroom	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	16	8.25	16	8.25
2	68	35.05	84	43.30
3	106	54.64	190	97.94
4	4	2.06	194	100.00

Frequency Table: Number of Bathrooms

The FREQ Procedure

Number of Bathrooms				
Bathroom	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	89	45.88	89	45.88
2	90	46.39	179	92.27
3	15	7.73	194	100.00

Frequency Table: Garage Type

The FREQ Procedure

Garage Type				
GarageType_Clean	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Attached	81	41.75	81	41.75
Detached	51	26.29	132	68.04
No garage/NA	21	10.82	153	78.87
Unknown	41	21.13	194	100.00

Frequency Table: Garage Status

The FREQ Procedure

Garage Status				
Garage_Status	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Garage	173	89.18	173	89.18
No garage	21	10.82	194	100.00

Frequency Table: Garage Type

The FREQ Procedure

Garage Type				
GarageType_Clean	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Attached	81	41.75	81	41.75
Detached	51	26.29	132	68.04
No garage/NA	21	10.82	153	78.87
Unknown	41	21.13	194	100.00

Bathrooms were mainly concentrated around **1** or **2 bathrooms**. The frequency table showed that **89 houses** had 1 bathroom, **90 houses** had 2 bathrooms, and **15 houses** had 3 bathrooms.

The average age of the houses when they were sold was **46.44 years**. The newest house was only **1 year old**, while the oldest house was **135 years old**. This shows that the dataset contains both newer and older houses.

Descriptive Statistics for Cleaned Housing Dataset

The MEANS Procedure

Variable	Label	N	Mean	Median	Std Dev	Minimum	Maximum
SoldPrice	Sale Price (£)	194	106155.08	104800.00	28938.35	28000.00	176000.00
LivingArea_sm	Living Area (Square Metres)	194	103.03	102.50	22.01	41.00	138.00
Garage_sm	Garage Area (Square Metres)	194	33.83	35.50	16.88	0.00	84.00
Bedroom	Number of Bedrooms	194	2.51	3.00	0.68	1.00	4.00
Bathroom	Number of Bathrooms	194	1.62	2.00	0.63	1.00	3.00
ConstructionYear	Construction Year	194	1961.47	1963.00	27.19	1875.00	2007.00
YearSold	Year Sold	194	2007.92	2008.00	1.33	2006.00	2010.00
HouseAge	House Age at Sale	194	46.44	45.50	27.13	1.00	135.00

Garage type clearly affected average sale price. Houses with **attached garages** had the highest average sale price at **£123,062.96**. Houses with **detached garages** had an average sale price of **£93,299.61**. **No garage/NA** had a lower average sale price of **£74,934.29**. Unknown garage type averaged **£104,733.80**.

Sale Price by Garage Type

The MEANS Procedure

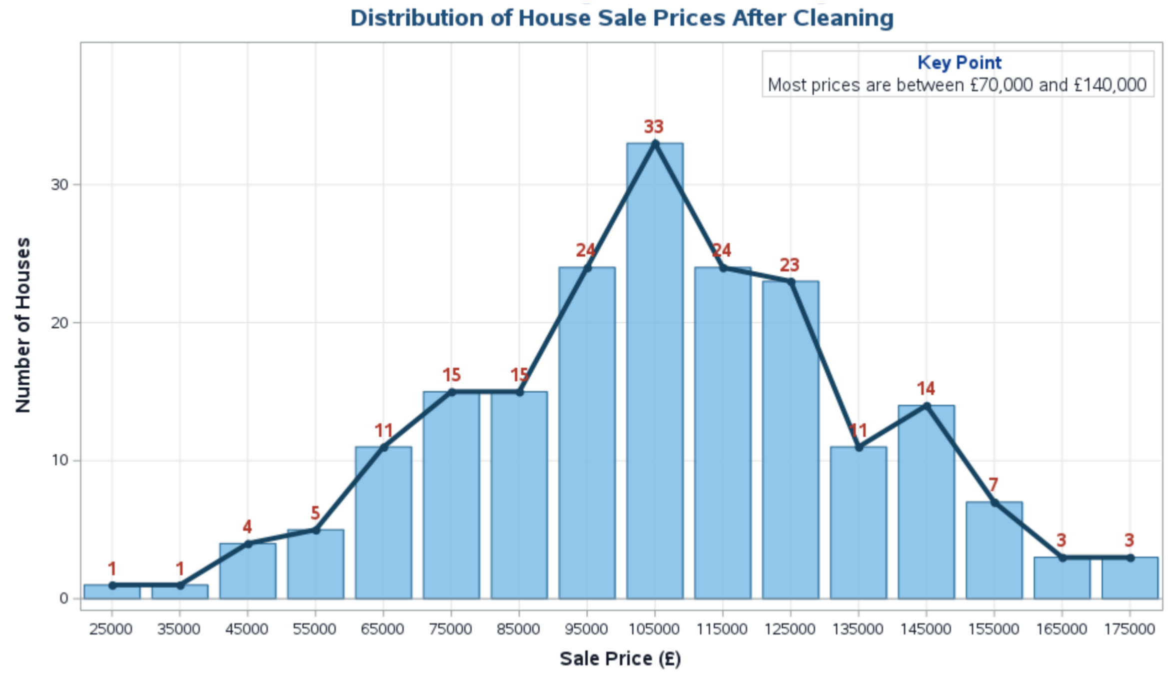
Analysis Variable : SoldPrice Sale Price (£)							
Garage Type	N Obs	N	Mean	Median	Std Dev	Minimum	Maximum
Attached	81	81	123062.96	124000.00	23212.65	65040.00	174400.00
Detached	51	51	93299.61	96600.00	23906.18	28000.00	137520.00
No garage/NA	21	21	74934.29	73520.00	20884.19	41600.00	116800.00
Unknown	41	41	104733.80	103600.00	26964.10	60000.00	176000.00

This indicates that **garage type is associated with sale price**. Homes with attached garages sell for more, while homes without garage information or without garage space sell for less.

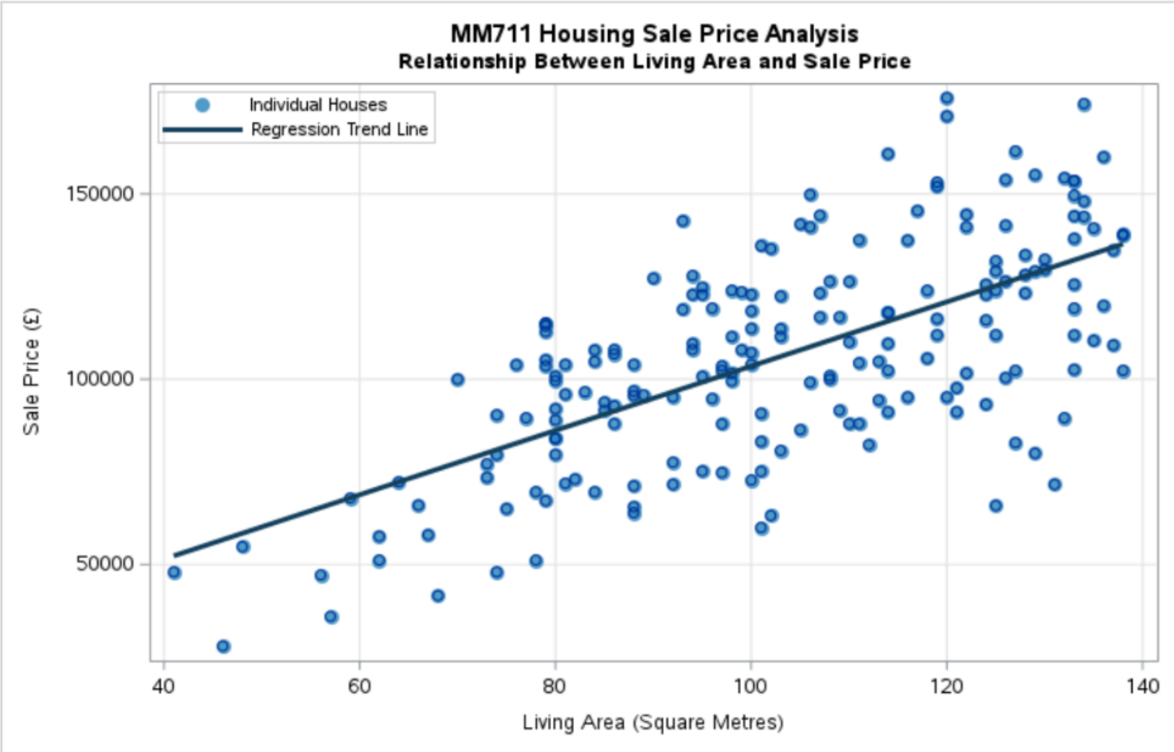
4. Visual Analysis

Several graphs were used to explore the data.

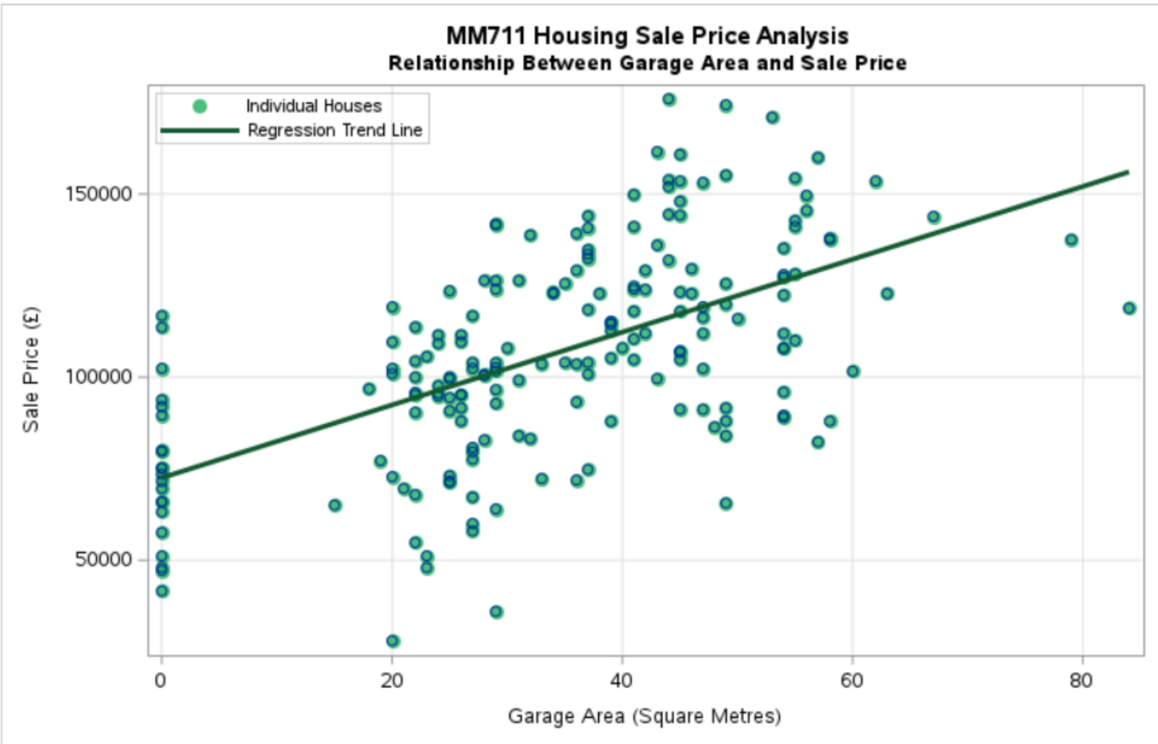
The outlier boxplots before cleaning showed which records needed attention. There was one extreme value in the living area boxplot and one extreme value in the garage area boxplot. These both related to reference **27183**, which had **1330 m^2** of living area and **380 m^2** of garage area. These values did not make logical sense compared with the rest of the dataset.



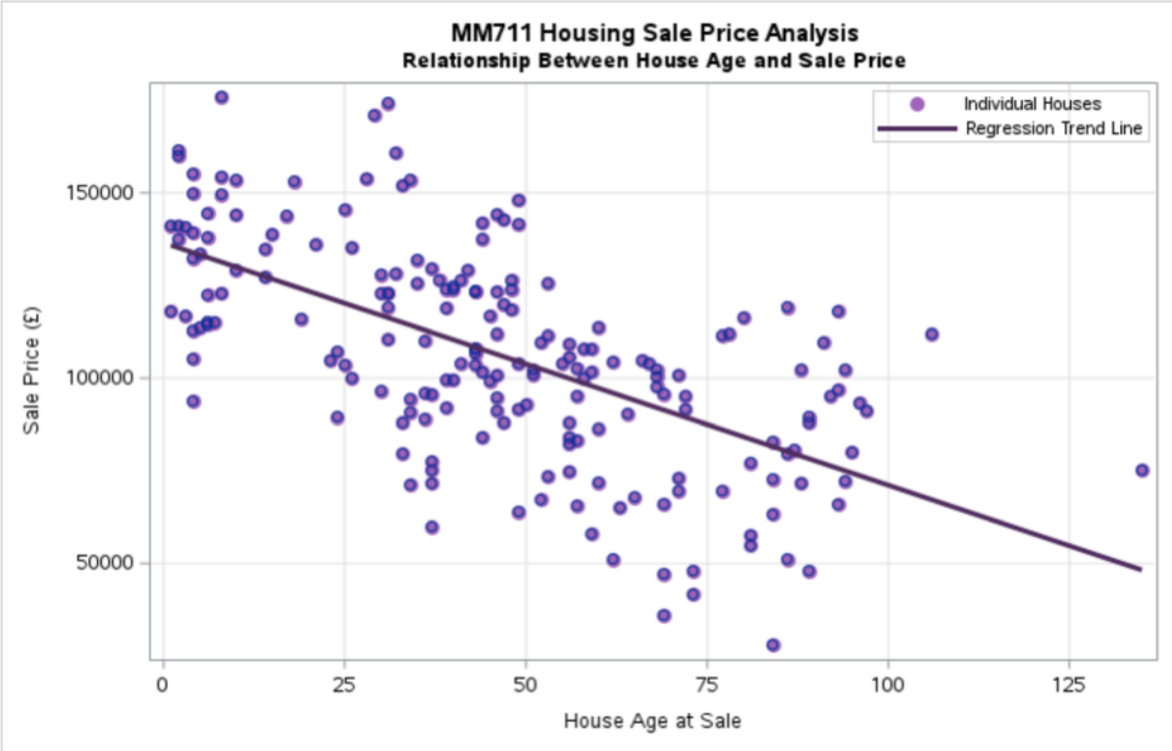
The histogram of sale price showed that most houses sold between about **£70,000** and **£140,000**. There were a few very low and high sale prices, but the distribution was still suitable for regression analysis after cleaning.



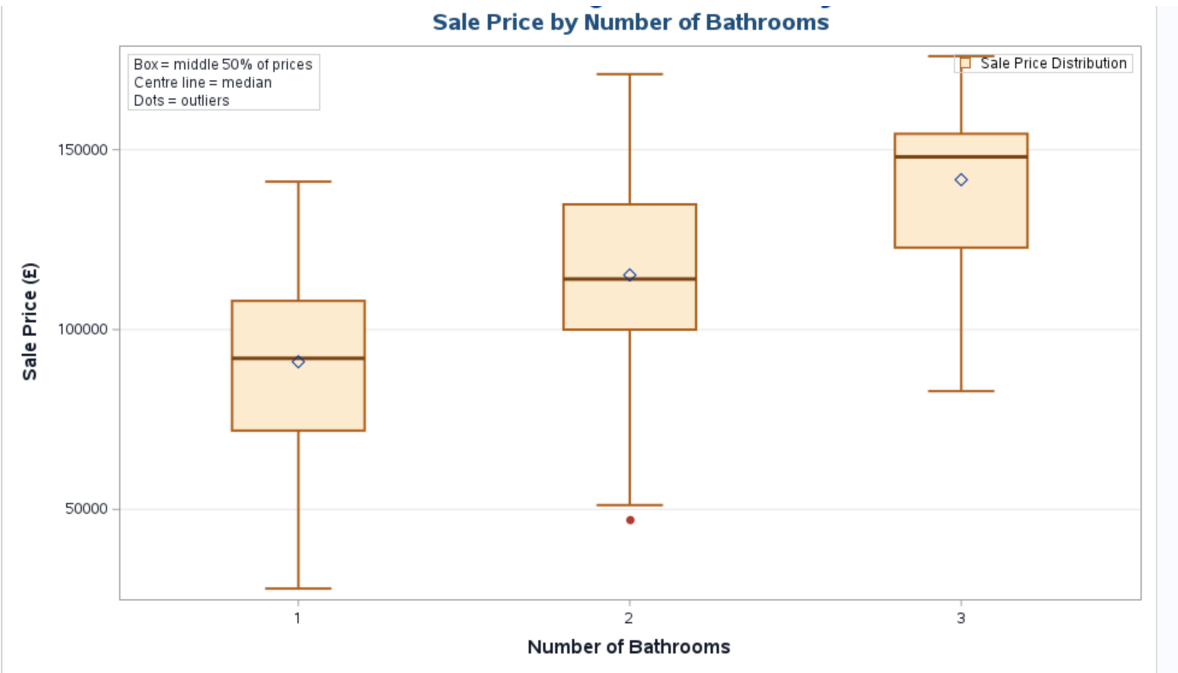
The relationship between **sale price** and **living area** was positively linear. This means that as living area increased, sale price also tended to increase. This was the clearest trend in the visual analysis.



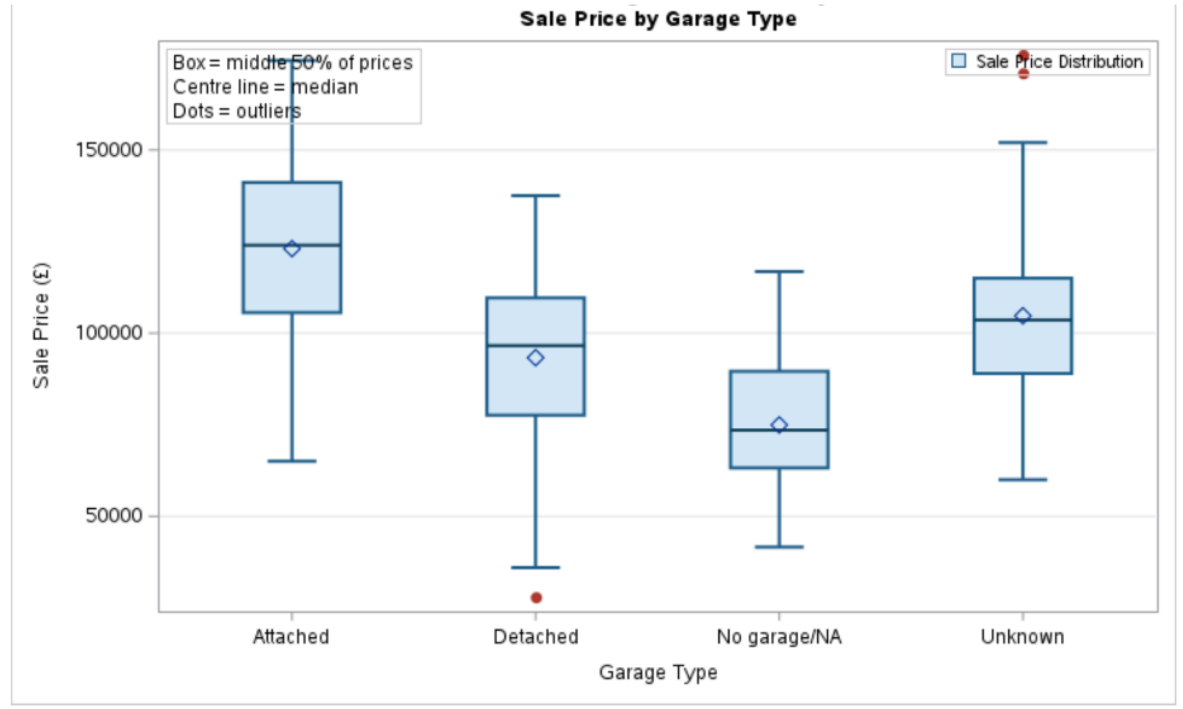
The scatter plot of sale price against garage area also showed a **positive relationship**. Houses with larger garage areas tended to have higher sale prices. This pattern was not as strong as living area, but it was still visible.



The scatter plot of sale price against house age indicated a **negative relationship**. Older houses tended to sell for lower prices. This is consistent with the correlation and regression results.



The distributions by number of bathrooms had similar shapes. As the number of bathrooms increased, sale price also tended to increase.



The visual results were consistent with the statistical results. **Living area**, **garage area**, **number of bathrooms**, **garage type**, and **house age** all appeared to **influence sale price**.

5. Correlation Analysis

Correlation analysis was used to measure the strength and direction of the relationship between sale price and the numerical variables.

Table 1: Correlation with SoldPrice

Variable	Correlation with SoldPrice	p-value
LivingArea_sm	**0.65981**	< 0.0001
Garage_sm	**0.58022**	< 0.0001
Bathroom	**0.53629**	< 0.0001
Bedroom	**0.25139**	**0.0004**
HouseAge	**-0.61332**	< 0.0001
YearSold	**0.09371**	**0.1937**

Correlation Between Sale Price and Numeric Variables

The CORR Procedure

7 Variables:	SoldPrice LivingArea_sm Garage_sm Bedroom Bathroom HouseAge YearSold

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum	Label
SoldPrice	194	106155	28938	20594086	28000	176000	Sale Price (£)
LivingArea_sm	194	103.02577	22.00552	19987	41.00000	138.00000	Living Area (Square Metres)
Garage_sm	194	33.82990	16.88108	6563	0	84.00000	Garage Area (Square Metres)
Bedroom	194	2.50515	0.67714	486.00000	1.00000	4.00000	Number of Bedrooms
Bathroom	194	1.61856	0.62658	314.00000	1.00000	3.00000	Number of Bathrooms
HouseAge	194	46.44330	27.12587	9010	1.00000	135.00000	House Age at Sale
YearSold	194	2008	1.33250	389536	2006	2010	Year Sold

Pearson Correlation Coefficients, N = 194
Prob > |r| under H0: Rho=0

	SoldPrice	LivingArea_sm	Garage_sm	Bedroom	Bathroom	HouseAge	YearSold
SoldPrice Sale Price (£)	1.00000	0.65981 <.0001	0.58022 <.0001	0.25139 0.0004	0.53629 <.0001	-0.61332 <.0001	0.09371 0.1937
LivingArea_sm Living Area (Square Metres)	0.65981 <.0001	1.00000	0.35926 <.0001	0.54052 <.0001	0.35094 <.0001	-0.20984 0.0033	0.01845 0.7985
Garage_sm Garage Area (Square Metres)	0.58022 <.0001	0.35926 <.0001	1.00000	0.09866 0.1711	0.33428 <.0001	-0.38148 <.0001	-0.03357 0.6422
Bedroom Number of Bedrooms	0.25139 0.0004	0.54052 <.0001	0.09866 0.1711	1.00000	0.07793 0.2801	0.03203 0.6575	0.00622 0.9314
Bathroom Number of Bathrooms	0.53629 <.0001	0.35094 <.0001	0.33428 <.0001	0.07793 0.2801	1.00000	-0.47775 <.0001	0.08624 0.2318
HouseAge House Age at Sale	-0.61332 <.0001	-0.20984 0.0033	-0.38148 <.0001	0.03203 0.6575	-0.47775 <.0001	1.00000	-0.02292 0.7511
YearSold Year Sold	0.09371 0.1937	0.01845 0.7985	-0.03357 0.6422	0.00622 0.9314	0.08624 0.2318	-0.02292 0.7511	1.00000

Living area showed the strongest positive correlation with sale price, **0.65981**, with a p-value below 0.0001. This means larger houses were strongly associated with higher sale prices.

Garage area also had a positive correlation with sale price, **0.58022**, and it was highly statistically significant at $p < 0.0001$.

Bathroom count had a positive correlation of **0.53629**, which means houses with more bathrooms were generally more expensive.

House age had a strong negative correlation of **-0.61332**. This means older houses tended to sell for lower prices.

Bedroom count had a weaker positive correlation of **0.25139**. It was statistically significant, but it was weaker than living area, garage area, bathroom count and house age.

YearSold had a weak positive correlation with sale price, but it was not statistically significant because the p-value was **0.1937**, which is above **0.05**. This means YearSold alone does not show a strong direct relationship with sale price.

6. Regression Analysis

A multiple regression model was used to estimate how each variable affected sale price, while holding all other variables constant. The primary regression model used was:

SoldPrice = *LivingArea_sm + Garage_sm + GarageType_Clean + Bedroom + Bathroom + HouseAge + YearSold*

PROC GLM was used because **GarageType_Clean** is a categorical variable. This is more appropriate than only using PROC REG, which is mainly used for numeric predictors.

The main regression model used **194 observations**. The model was statistically significant, with an F value of **61.59** and $p < 0.0001$.

The R-square value was **0.750772**. This means the model explained approximately **75.1%** of the variation in sale price, which is a strong result for this housing dataset.

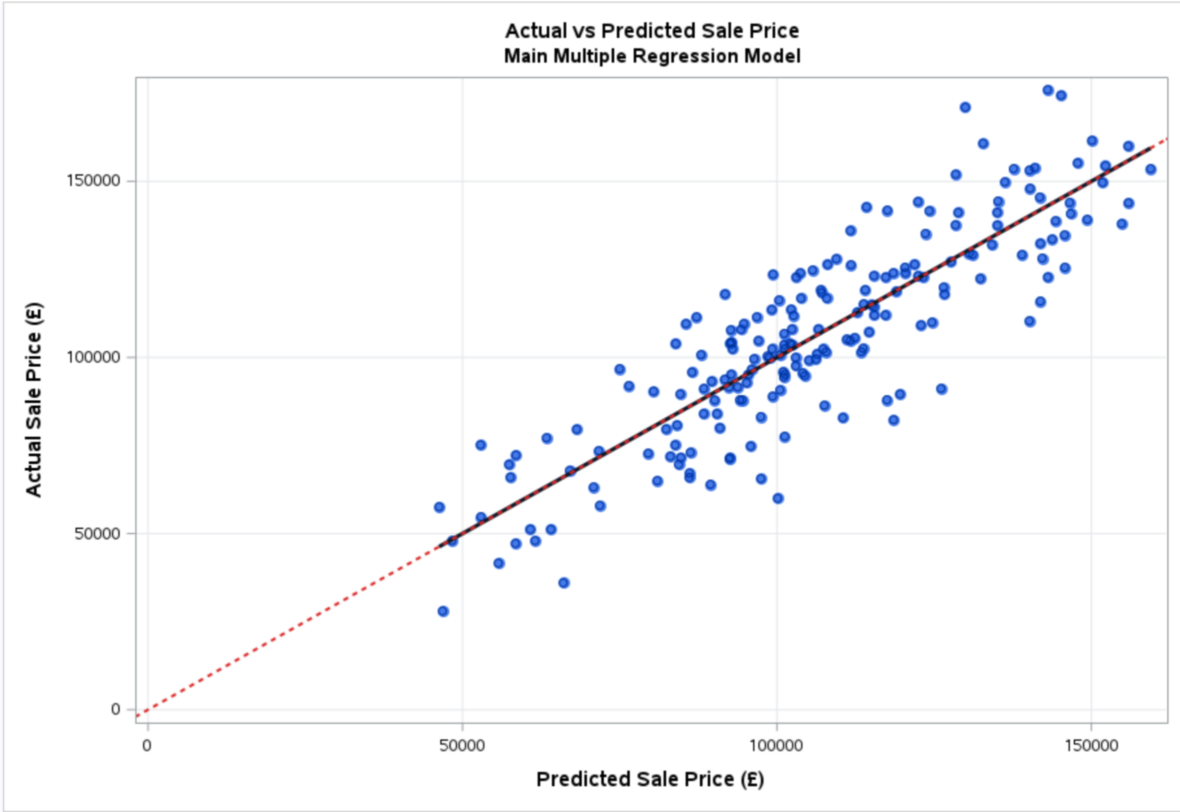
Main Multiple Regression Model for House Sale Price

The GLM Procedure

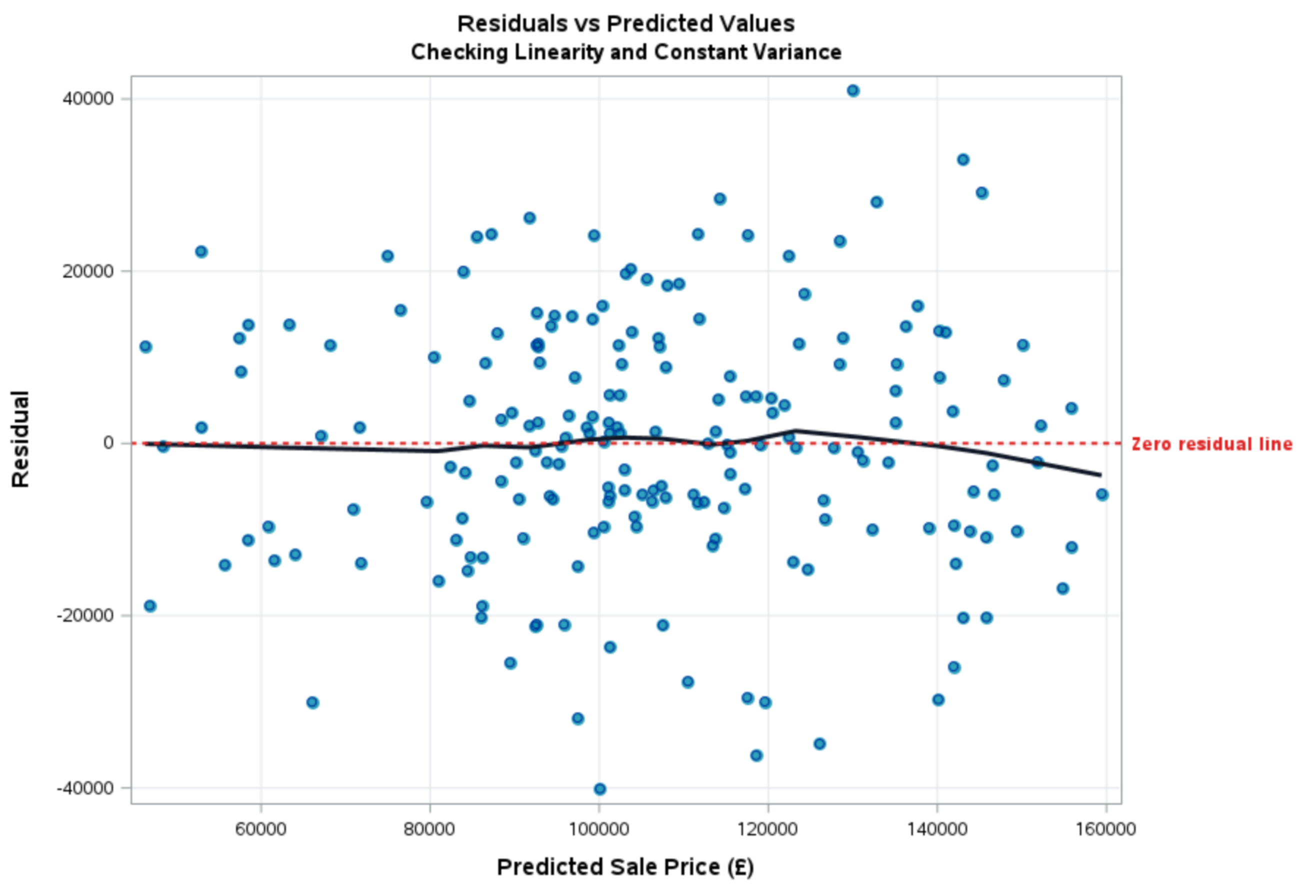
Dependent Variable: SoldPrice Sale Price (£)

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	121342566362	13482507374	61.59	<.0001
Error	184	40281073550	218918877.99		
Corrected Total	193	161623639912			

R-Square	Coeff Var	Root MSE	SoldPrice Mean
0.750772	13.93801	14795.91	106155.1



Actual vs Predicted Sale Price
Main Multiple Regression Model
Actual Sale Price (£)
Predicted Sale Price (£)
0
50000
100000
150000



Residuals vs Predicted Values
Checking Linearity and Constant Variance
Residual
Predicted Sale Price (£)
Zero residual line
40000
20000
0
-20000
-40000
60000
80000
100000
120000
140000
160000

Parameter	Estimate		Standard Error	t Value	Pr > |t|
Intercept	-3309936.434	B	1615986.484	-2.05	0.0420
LivingArea_sm	555.725		67.521	8.23	<.0001
Garage_sm	328.490		100.681	3.26	0.0013
GarageType_Clean Attached	4780.328	B	2986.738	1.60	0.1112
GarageType_Clean Detached	-4387.593	B	3222.840	-1.36	0.1751
GarageType_Clean No garage/NA	-7418.750	B	5373.859	-1.38	0.1691
GarageType_Clean Unknown	0.000	B	.	.	.
Bedroom	-840.939		1910.679	-0.44	0.6604
Bathroom	6356.755		2106.388	3.02	0.0029
HouseAge	-341.417		50.137	-6.81	<.0001
YearSold	1671.065		804.938	2.08	0.0393

Table 2: Regression Interpretation Table

Variable	Estimate	p-value	Interpretation
LivingArea_sm	555.725	< 0.0001	Each extra square metre of living area increased predicted sale price by about £556
Garage_sm	328.490	0.0013	Each extra square metre of garage area increased predicted sale price by about £328
Bedroom	-840.939	0.6604	Bedroom count was not statistically significant after other variables were included
Bathroom	6356.755	0.0029	Each extra bathroom increased predicted sale price by about £6,357
HouseAge	-341.417	< 0.0001	Each extra year of house age reduced predicted sale price by about £341
YearSold	1671.065	0.0393	Later sale years were linked with higher predicted prices

Living area was the strongest predictor in the model. For every additional square metre of living area, a house was predicted to achieve £556 higher sale price, all other variables being equal.

Garage area was also significant. The estimate was 328.490, meaning that each additional square metre of garage area increased predicted sale price by about £328.

Bathroom count was significant. Each additional bathroom increased the predicted sale price by about £6,357. This shows that bathrooms add value to a house.

House age had a significant negative effect. Each extra year of age reduced sale price by about £341 on average, suggesting that newer houses were generally more valuable in this dataset.

YearSold had a p-value of 0.0393. Holding all other features constant, later sale years were linked with slightly higher predicted prices.

Bedroom count was not statistically significant in the regression model, with a p-value of 0.6604. This does not mean bedrooms have no value. It means that after living area, bathrooms, garage area, house age and year sold were known, bedroom count did not add much more information in predicting price.

Garage type was significant overall. The Type III test for **GarageType_Clean** had a p-value of **0.0091**. This tells us that garage type had an effect on sale price after controlling for the other variables in the model. The model output also gave a caution that the model was **singular** and used a **generalized inverse**. Because of this, it is safer to focus on the overall Type III p-value when deciding whether garage type matters.

In this model, **GarageType_Clean Unknown** acts as the reference category (**estimate = 0**). The coefficients for **Attached**, **Detached**, and **No garage/NA** are therefore interpreted relative to **Unknown** garage type.

7. Alternative and Robustness Checks

An alternative regression model was also run without garage type. This model used only numeric predictors. It had an R-square value of **0.7347** and an adjusted R-square value of **0.7261**. This means the numeric-only model explained about **73.5%** of the variation in sale price.

Alternative Regression Model Without Garage Type

The REG Procedure
Model: MODEL1
Dependent Variable: SoldPrice Sale Price (£)

Number of Observations Read	194
Number of Observations Used	194

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	1.187386E11	19789763493	86.29	<.0001
Error	187	42885058955	229331866		
Corrected Total	193	1.616236E11			

Root MSE	15144	R-Square	0.7347
Dependent Mean	106155	Adj R-Sq	0.7261
Coeff Var	14.26565		

Parameter Estimates								
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	Intercept	1	-3237506	1652257	-1.96	0.0515	.	0
LivingArea_sm	Living Area (Square Metres)	1	616.16336	65.98247	9.34	<.0001	0.56362	1.77424
Garage_sm	Garage Area (Square Metres)	1	409.69025	74.14717	5.53	<.0001	0.75843	1.31851
Bedroom	Number of Bedrooms	1	-974.37855	1952.65883	-0.50	0.6184	0.67966	1.47132
Bathroom	Number of Bathrooms	1	5143.28009	2095.64534	2.45	0.0150	0.68915	1.45106
HouseAge	House Age at Sale	1	-392.76655	47.77572	-8.22	<.0001	0.70750	1.41343

Parameter Estimates								
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
YearSold	Year Sold	1	1632.87476	822.96126	1.98	0.0487	0.98814	1.01201

The main model with garage type had a higher R-square value of **0.7508**. This shows that including garage type improved the model. Therefore, the PROC GLM model with garage type is the better main model for the report.

The numeric-only regression also included tolerance and variance inflation factor values. The VIF values were all below **2**. This suggests that multicollinearity was not a serious problem in the numeric-only model.

Robustness Check: Regression Model Using Log Sale Price

The GLM Procedure

Dependent Variable: LogSoldPrice Log of Sale Price

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	13.19781085	1.46642343	54.07	<.0001
Error	184	4.99025170	0.02712093		
Corrected Total	193	18.18806255			

R-Square	Coeff Var	Root MSE	LogSoldPrice Mean
0.725630	1.428298	0.164684	11.53011

A log-transformed sale price model was also run as a robustness check. This model used **LogSoldPrice** as the dependent variable. It had an R-square value of **0.725630** and was statistically significant with $p < 0.0001$.

Source	DF	Type III SS	Mean Square	F Value	Pr > F
LivingArea_sm	1	1.65466300	1.65466300	61.01	<.0001
Garage_sm	1	0.19130439	0.19130439	7.05	0.0086
GarageType_Clean	3	0.27581481	0.09193827	3.39	0.0192
Bedroom	1	0.03670696	0.03670696	1.35	0.2462
Bathroom	1	0.12519654	0.12519654	4.62	0.0330
HouseAge	1	1.15363337	1.15363337	42.54	<.0001
YearSold	1	0.09275605	0.09275605	3.42	0.0660

The log model showed the same general pattern as the main model. Living area, garage area, garage type, bathroom count and house age remained important. YearSold became weaker in the log model because its p-value was **0.0660**. This supports the reliability of the main findings.

8. Discussion

The results show that house sale price is mainly influenced by **property size**, **garage area**, **number of bathroo** **garage type**, and **house age**.

The most important positive predictor was **living area**. This is logical because larger houses provide more usable living space. Therefore, larger houses are expected to have higher sale prices.

Garage area also had a positive effect, suggesting that garage space adds value to a property for parking, storage or convenience. The grouped summary also showed that houses with garages had a much higher mean sale price than houses with no garage recorded.

Bathroom count was another important factor. Houses with **3 bathrooms** had a mean sale price of **£141,656**, compared with **£91,051.24** for houses with **1 bathroom**. This shows that bathrooms are one of the more valuable property features.

Sale Price by Number of Bathrooms

The MEANS Procedure

Analysis Variable : SoldPrice Sale Price (£)							
Number of Bathrooms	N Obs	N	Mean	Median	Std Dev	Minimum	Maximum
1	89	89	91051.24	92000.00	24493.74	28000.00	141120.00
2	90	90	115174.28	114050.00	24547.01	47200.00	171000.00
3	15	15	141656.00	148000.00	24888.43	82880.00	176000.00

House age had a strong negative effect. Older houses tended to sell for less. This may be because older houses often need more repairs, have older layouts or require updating.

Bedroom count was weaker than expected. Although it had a positive correlation with sale price, it was not significant in the regression model. This is likely because bedroom count overlaps with living area. Once living area is included, bedroom count adds less new information.

Garage type also made a difference. **Attached garages** had the highest average sale price, while detached garages and no garage information were linked with lower prices. This may reflect buyer preference, convenience or differences in the type and quality of houses with attached garages.

Overall, the regression model explained approximately **75%** of the sale price variation. This shows that the selected variables explain a large part of house price differences in this dataset.

9. Limitations

There are several limitations.

First, the dataset had **missing values**, **duplicate records**, **invalid values** and **outliers**. These were treated before analysis, but different cleaning decisions could slightly change the results.

Second, the dataset did not include **location**. Location is often one of the most important factors affecting house price, so its absence limits the model.

Third, the dataset did not include land size, property condition, neighbourhood quality, school quality, distance to services or local market demand. These factors may also influence sale price.

Fourth, **Garage_Type** had blank values and NA values. These were cleaned into **Unknown** and **No garage/NA**, but there may still be some mislabelling.

Finally, the extreme outlier at reference **27183** was removed because it appeared unrealistic. If the record was actually correct, deleting it would exclude one premium large property. Based on the remaining data, it was more practical to treat it as a data entry error.

Lastly, the regression model shows relationship, not cause and effect. For example, the model shows that higher sale price is linked with larger living area, but this does not prove that size alone caused the price difference.

10. Conclusion

This report examined the impact of housing characteristics on sale price using two linked datasets. The data was imported, checked, merged, cleaned, explored and visualized before correlation and regression analysis were applied.

The final dataset contained **194 houses**. The main regression model explained approximately **75.1%** of the variation in sale price.

The results showed that houses with larger living areas sold at higher prices. Bigger garage area also increased predicted sale price. A higher number of bathrooms had a positive relationship with sale price. Older houses generally sold for lower prices, and garage type clearly affected average sale price, with attached garages having the highest average prices.

Bedroom count was less important after controlling for living area and other variables. This suggests that total living space is a better predictor of sale price than bedroom count alone.

The main findings are that **property size**, **garage area**, **bathroom count**, **garage type**, and **house age** are useful predictors of sale price in this dataset. More features such as **location**, **property condition**, **land size**, and **neighbourhood details** would make the analysis stronger.

References

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A. SAS Code

```
  1. LIBRARY, OUTPUT AND GRAPH SETTINGS

libname MM711 "/home/u64455556/MM711/hpd18 MYDATA";

options nodate nonumber;

ods listing gpath="/home/u64455556/MM711/hpd18 MYDATA/report_images";

ods html path="/home/u64455556/MM711/hpd18 MYDATA/report_images"
         file="fixed_report_outputs.html"
         style=HTMLBlue;

ods graphics on / reset=all
    width=10in
    height=6.2in
    imagefmt=png
    scale=on
    imagemap;

title;
footnote;


  2. IMPORT SOURCE DATA


proc import datafile="/home/u64455556/MM711/hpd18 MYDATA/Characteristics.xlsx"
    out=characteristics
    dbms=xlsx
    replace;
    getnames=yes;
run;

proc import datafile="/home/u64455556/MM711/hpd18 MYDATA/Houseprice.xlsx"
    out=houseprice
    dbms=xlsx
    replace;
    getnames=yes;
run;


  3. ADD VARIABLE LABELS


data characteristics;
    set characteristics;

    label
        Reference     = "House Reference Number"
        LivingArea_sm = "Living Area (Square Metres)"
        Garage_sm     = "Garage Area (Square Metres)"
        Garage_Type   = "Garage Type"
        Bedroom       = "Number of Bedrooms"
        Bathroom      = "Number of Bathrooms";
run;

data houseprice;
    set houseprice;

    format SoldPrice pound12.2;

    label
        Reference        = "House Reference Number"
```

```
        ConstructionYear = "Construction Year"
        YearSold         = "Year Sold"
        SoldPrice        = "Sale Price (  )";
run;


  4. SORT DATA AND REMOVE DUPLICATES


proc sort data=characteristics
    out=characteristics_sorted
    nodupkey
    dupout=characteristics_duplicates;
    by Reference;
run;

proc sort data=houseprice
    out=houseprice_sorted
    nodupkey
    dupout=houseprice_duplicates;
    by Reference;
run;


  5. MERGE DATASETS


data housing_raw;
    merge houseprice_sorted(in=a)
          characteristics_sorted(in=b);
    by Reference;

    if a and b;

    format SoldPrice pound12.2;

    label
        Reference        = "House Reference Number"
        ConstructionYear = "Construction Year"
        YearSold         = "Year Sold"
        SoldPrice        = "Sale Price (  )"
        LivingArea_sm    = "Living Area (Square Metres)"
        Garage_sm        = "Garage Area (Square Metres)"
        Garage_Type      = "Garage Type"
        Bedroom          = "Number of Bedrooms"
        Bathroom         = "Number of Bathrooms";
run;


  6. CLEAN DATA


data housing_clean;
    set housing_raw;

    length GarageType_Clean $20;
    length Garage_Status    $15;

    HouseAge = YearSold - ConstructionYear;

    if missing(Garage_Type) then
        GarageType_Clean = "Unknown";
    else if upcase(strip(Garage_Type)) = "NA" and Garage_sm = 0 then
        GarageType_Clean = "No garage/NA";
    else
        GarageType_Clean = strip(Garage_Type);

```

```
    if Garage_sm > 0 then
        Garage_Status = "Garage";
    else
        Garage_Status = "No garage";

    if missing(SoldPrice)        then delete;
    if missing(YearSold)         then delete;
    if missing(ConstructionYear) then delete;

    if LivingArea_sm <= 0 then delete;
    if Garage_sm < 0      then delete;
    if HouseAge < 0       then delete;

    if Reference = 27183 then delete;

    format SoldPrice pound12.2;

    label
        Reference        = "House Reference Number"
        SoldPrice        = "Sale Price (  )"
        LivingArea_sm    = "Living Area (Square Metres)"
        Garage_sm        = "Garage Area (Square Metres)"
        GarageType_Clean = "Garage Type"
        Garage_Status    = "Garage Status"
        Bedroom          = "Number of Bedrooms"
        Bathroom         = "Number of Bathrooms"
        ConstructionYear = "Construction Year"
        YearSold         = "Year Sold"
        HouseAge         = "House Age at Sale";
run;


  7. FREQUENCY TABLES


title1 "MM711 Housing Sale Price Analysis";
title2 "Frequency Table: Number of Bedrooms";

proc freq data=housing_clean;
    tables Bedroom / missing;
run;

title2 "Frequency Table: Number of Bathrooms";

proc freq data=housing_clean;
    tables Bathroom / missing;
run;

title2 "Frequency Table: Garage Type";

proc freq data=housing_clean;
    tables GarageType_Clean / missing;
run;


  8. DESCRIPTIVE STATISTICS


title2 "Descriptive Statistics for Cleaned Housing Dataset";

proc means data=housing_clean n mean median std min max maxdec=2;
    var SoldPrice LivingArea_sm Garage_sm Bedroom
        Bathroom ConstructionYear YearSold HouseAge;
    format SoldPrice pound12.2;
run;

title2 "Sale Price by Garage Type";
```

```
proc means data=housing_clean n mean median std min max maxdec=2;
    class GarageType_Clean;
    var SoldPrice;
    format SoldPrice pound12.2;
run;

title2 "Sale Price by Number of Bathrooms";

proc means data=housing_clean n mean median std min max maxdec=2;
    class Bathroom;
    var SoldPrice;
    format SoldPrice pound12.2;
run;


  9. HIGH-QUALITY HISTOGRAM


data housing_hist;
    set housing_clean;

    BinWidth = 10000;
    BinLow   = floor(SoldPrice / BinWidth) * BinWidth;
    BinHigh  = BinLow + BinWidth;
    BinMid   = BinLow + (BinWidth / 2);

    PriceRange = cats(put(BinLow, pound8.0), " - ", put(BinHigh, pound8.0));
run;

proc summary data=housing_hist nway;
    class BinLow BinHigh BinMid PriceRange;
    output out=hist_counts(drop=_type_ rename=(_freq_=Frequency));
run;

data hist_counts;
    set hist_counts;

    LabelY         = Frequency + 1.2;
    FrequencyLabel = strip(put(Frequency, 8.));
run;

ods graphics / imagename="sale_price_histogram_high_quality";

title1 height=16pt bold color=CX0B3D91
    "MM711 Housing Sale Price Analysis";

title2 height=12pt color=CX1F4E79
    "Distribution of House Sale Prices After Cleaning";

proc sgplot data=hist_counts noautolegend;

    vbarparm category=BinMid response=Frequency /
        barwidth=0.82
        fillattrs=(color=CX85C1E9 transparency=0.10)
        outlineattrs=(color=CX21618C thickness=1.5);

    series x=BinMid y=Frequency /
        lineattrs=(color=CX154360 thickness=4)
        markers
        markerattrs=(symbol=circlefilled color=CX154360 size=8);

    text x=BinMid y=LabelY text=FrequencyLabel /
        strip
        textattrs=(size=10 weight=bold color=CXB03A2E);

    inset "Most prices are between  70 ,000 and  140 ,000" /
```

```
        position=topright
        border
        title="Key Point"
        titleattrs=(size=10 weight=bold color=CX0B3D91)
        textattrs=(size=10 color=CX111827);

    xaxis label="Sale Price (  )"
        grid
        valuesformat=pound12.0
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827)
        offsetmin=0.03
        offsetmax=0.03;

    yaxis label="Number of Houses"
        grid
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827)
        offsetmax=0.15;

    format BinMid pound12.0;
run;


  10. LIVING AREA VS SALE PRICE


ods graphics / imagename="living_area_sale_price_high_quality";

title1 height=16pt bold color=CX0B3D91
    "MM711 Housing Sale Price Analysis";

title2 height=12pt color=CX1F4E79
    "Relationship Between Living Area and Sale Price";

proc sgplot data=housing_clean;

    scatter x=LivingArea_sm y=SoldPrice /
        name="points"
        legendlabel="Individual Houses"
        markerattrs=(symbol=circlefilled color=CX2E86C1 size=9)
        transparency=0.18;

    reg x=LivingArea_sm y=SoldPrice /
        name="trend"
        legendlabel="Regression Trend Line"
        lineattrs=(color=CX154360 thickness=4);

    keylegend "points" "trend" /
        location=inside
        position=topleft
        across=1
        valueattrs=(size=9);

    xaxis label="Living Area (Square Metres)"
        grid
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    yaxis label="Sale Price (  )"
        grid
        valuesformat=pound12.0
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    format SoldPrice pound12.0;
run;
```

```
  11. GARAGE AREA VS SALE PRICE


ods graphics / imagename="garage_area_sale_price_high_quality";

title1 height=16pt bold color=CX0B3D91
    "MM711 Housing Sale Price Analysis";

title2 height=12pt color=CX1F4E79
    "Relationship Between Garage Area and Sale Price";

proc sgplot data=housing_clean;

    scatter x=Garage_sm y=SoldPrice /
        name="points"
        legendlabel="Individual Houses"
        markerattrs=(symbol=circlefilled color=CX28B463 size=9)
        transparency=0.18;

    reg x=Garage_sm y=SoldPrice /
        name="trend"
        legendlabel="Regression Trend Line"
        lineattrs=(color=CX145A32 thickness=4);

    keylegend "points" "trend" /
        location=inside
        position=topleft
        across=1
        valueattrs=(size=9);

    xaxis label="Garage Area (Square Metres)"
        grid
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    yaxis label="Sale Price (  )"
        grid
        valuesformat=pound12.0
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    format SoldPrice pound12.0;
run;


  12. HOUSE AGE VS SALE PRICE


ods graphics / imagename="house_age_sale_price_high_quality";

title1 height=16pt bold color=CX0B3D91
    "MM711 Housing Sale Price Analysis";

title2 height=12pt color=CX1F4E79
    "Relationship Between House Age and Sale Price";

proc sgplot data=housing_clean;

    scatter x=HouseAge y=SoldPrice /
        name="points"
        legendlabel="Individual Houses"
        markerattrs=(symbol=circlefilled color=CX8E44AD size=9)
        transparency=0.18;

    reg x=HouseAge y=SoldPrice /
        name="trend"
```

```
        legendlabel="Regression Trend Line"
        lineattrs=(color=CX512E5F thickness=4);

    keylegend "points" "trend" /
        location=inside
        position=topright
        across=1
        valueattrs=(size=9);

    xaxis label="House Age at Sale"
        grid
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    yaxis label="Sale Price (  )"
        grid
        valuesformat=pound12.0
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    format SoldPrice pound12.0;
run;


  13. SALE PRICE BY NUMBER OF BATHROOMS


ods graphics / imagename="sale_price_by_bathrooms_high_quality";

title1 height=16pt bold color=CX0B3D91
    "MM711 Housing Sale Price Analysis";

title2 height=12pt color=CX1F4E79
    "Sale Price by Number of Bathrooms";

proc sgplot data=housing_clean;

    vbox SoldPrice / category=Bathroom
        name="box"
        legendlabel="Sale Price Distribution"
        fillattrs=(color=CXFDEBD0)
        lineattrs=(color=CXAF601A thickness=2)
        medianattrs=(color=CX784212 thickness=3)
        whiskerattrs=(color=CXAF601A thickness=2)
        outlierattrs=(color=CXB03A2E symbol=circlefilled size=7);

    keylegend "box" /
        location=inside
        position=topright
        valueattrs=(size=9);

    inset "Box = middle 50% of prices"
          "Centre line = median"
          "Dots = outliers" /
          position=topleft
          border
          textattrs=(size=9 color=CX111827);

    xaxis label="Number of Bathrooms"
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    yaxis label="Sale Price (  )"
        grid
        valuesformat=pound12.0
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);
```

```

    format SoldPrice pound12.0;
run;


  14. SALE PRICE BY NUMBER OF BEDROOMS


ods graphics / imagename="sale_price_by_bedrooms_high_quality";

title1 height=16pt bold color=CX0B3D91
    "MM711 Housing Sale Price Analysis";

title2 height=12pt color=CX1F4E79
    "Sale Price by Number of Bedrooms";

proc sgplot data=housing_clean;

    vbox SoldPrice / category=Bedroom
        name="box"
        legendlabel="Sale Price Distribution"
        fillattrs=(color=CXD5F5E3)
        lineattrs=(color=CX196F3D thickness=2)
        medianattrs=(color=CX145A32 thickness=3)
        whiskerattrs=(color=CX196F3D thickness=2)
        outlierattrs=(color=CXB03A2E symbol=circlefilled size=7);

    keylegend "box" /
        location=inside
        position=topright
        valueattrs=(size=9);

    inset "Box = middle 50% of prices"
          "Centre line = median"
          "Dots = outliers" /
          position=topleft
          border
          textattrs=(size=9 color=CX111827);

    xaxis label="Number of Bedrooms"
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    yaxis label="Sale Price (  )"
        grid
        valuesformat=pound12.0
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    format SoldPrice pound12.0;
run;


  15. SALE PRICE BY GARAGE TYPE


ods graphics / imagename="sale_price_by_garage_type_high_quality";

title1 height=16pt bold color=CX0B3D91
    "MM711 Housing Sale Price Analysis";

title2 height=12pt color=CX1F4E79
    "Sale Price by Garage Type";

proc sgplot data=housing_clean;

    vbox SoldPrice / category=GarageType_Clean
```

```
        name="box"
        legendlabel="Sale Price Distribution"
        fillattrs=(color=CXD6EAF8)
        lineattrs=(color=CX1F618D thickness=2)
        medianattrs=(color=CX154360 thickness=3)
        whiskerattrs=(color=CX1F618D thickness=2)
        outlierattrs=(color=CXB03A2E symbol=circlefilled size=7);

    keylegend "box" /
        location=inside
        position=topright
        valueattrs=(size=9);

    inset "Box = middle 50% of prices"
          "Centre line = median"
          "Dots = outliers" /
          position=topleft
          border
          textattrs=(size=9 color=CX111827);

    xaxis label="Garage Type"
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    yaxis label="Sale Price (  )"
        grid
        valuesformat=pound12.0
        valueattrs=(size=9 color=CX111827)
        labelattrs=(size=11 weight=bold color=CX111827);

    format SoldPrice pound12.0;
run;


  16. CORRELATION ANALYSIS


ods graphics / imagename="correlation_matrix_high_quality";

title1 height=16pt bold color=CX0B3D91
    "MM711 Housing Sale Price Analysis";

title2 height=12pt color=CX1F4E79
    "Correlation Between Sale Price and Numeric Variables";

proc corr data=housing_clean plots=matrix(histogram);
    var SoldPrice LivingArea_sm Garage_sm Bedroom
        Bathroom HouseAge YearSold;
run;

  17. MAIN MULTIPLE REGRESSION MODEL


ods graphics / imagename="main_regression_diagnostics_high_quality";

title1 "MM711 Housing Sale Price Analysis";
title2 "Main Multiple Regression Model for House Sale Price";

proc glm data=housing_clean plots=diagnostics;
    class GarageType_Clean;

    model SoldPrice =
        LivingArea_sm
        Garage_sm
        GarageType_Clean
        Bedroom
        Bathroom
```

```
        HouseAge
        YearSold
        / solution;
run;
quit;


  18. ALTERNATIVE NUMERIC-ONLY REGRESSION MODEL


ods graphics / imagename="numeric_only_regression_diagnostics_high_quality";

title1 "MM711 Housing Sale Price Analysis";
title2 "Alternative Regression Model Without Garage Type";

proc reg data=housing_clean plots=diagnostics;
    model SoldPrice =
        LivingArea_sm
        Garage_sm
        Bedroom
        Bathroom
        HouseAge
        YearSold
        / vif tol;
run;
quit;


  19. LOG-TRANSFORMED SALE PRICE MODEL


data housing_clean;
    set housing_clean;

    LogSoldPrice = log(SoldPrice);

    label LogSoldPrice = "Log of Sale Price";
run;

ods graphics / imagename="log_regression_diagnostics_high_quality";

title1 "MM711 Housing Sale Price Analysis";
title2 "Robustness Check: Regression Model Using Log Sale Price";

proc glm data=housing_clean plots=diagnostics;
    class GarageType_Clean;

    model LogSoldPrice =
        LivingArea_sm
        Garage_sm
        GarageType_Clean
        Bedroom
        Bathroom
        HouseAge
        YearSold
        / solution;
run;
quit;


  20. SAVE DIAGNOSTIC VALUES FROM MAIN REGRESSION MODEL


proc glm data=housing_clean plots=none;
    class GarageType_Clean;

    model SoldPrice =
```

```
        LivingArea_sm
        Garage_sm
        GarageType_Clean
        Bedroom
        Bathroom
        HouseAge
        YearSold
        / solution;

    output out=diag_soldprice
        predicted = Predicted_Value
        residual  = Residual
        student   = Student_Residual
        h         = Leverage
        cookd     = Cooks_D;
run;
quit;


  21. PREPARE DIAGNOSTIC DATASET


data diag_soldprice;
    set diag_soldprice;

    Observation           = _N_;
    Abs_Student_Residual  = abs(Student_Residual);

    label
        SoldPrice               = "Actual Sale Price (  )"
        Predicted_Value         = "Predicted Sale Price (  )"
        Residual                = "Residual"
        Student_Residual        = "Studentized Residual"
        Abs_Student_Residual    = "Absolute Studentized Residual"
        Leverage                = "Leverage"
        Cooks_D                 = "Cook's Distance"
        Observation             = "Observation Number";
run;


  22. CLEAN GRAPH STYLE FOR DIAGNOSTIC PLOTS


proc template;
    define style styles.MM711Clean;
        parent = styles.HTMLBlue;

        style GraphFonts /
            'GraphTitleFont'  = ("Arial", 15pt, bold)
            'GraphTitle1Font' = ("Arial", 13pt, bold)
            'GraphLabelFont'  = ("Arial", 11pt)
            'GraphValueFont'  = ("Arial", 9pt)
            'GraphDataFont'   = ("Arial", 9pt);

        style GraphWalls /
            frameborder     = on
            linestyle       = 1
            contrastcolor   = cxD1D5DB
            backgroundcolor = cxFFFFFF;

        style GraphBackground /
            backgroundcolor = cxFFFFFF;
    end;
run;

ods html style=MM711Clean;

```

```
ods graphics / reset
    width=9.5in
    height=6.5in
    imagefmt=png
    scale=on
    antialiasmax=5000
    attrpriority=none;


  23. DIAGNOSTIC GRAPH 1: ACTUAL VS PREDICTED SALE PRICE


ods graphics / imagename="01_actual_vs_predicted_sale_price";

title1 "Actual vs Predicted Sale Price";
title2 "Main Multiple Regression Model";

proc sgplot data=diag_soldprice noautolegend;

    scatter x=Predicted_Value y=SoldPrice /
        markerattrs=(symbol=circlefilled size=9 color=CX2563EB)
        transparency=0.15;

    reg x=Predicted_Value y=SoldPrice /
        lineattrs=(color=CX111827 thickness=3);

    lineparm x=0 y=0 slope=1 /
        lineattrs=(color=CXDC2626 thickness=2 pattern=shortdash);

    xaxis label="Predicted Sale Price (  )"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);

    yaxis label="Actual Sale Price (  )"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);

    format Predicted_Value SoldPrice pound12.0;
run;


  24. DIAGNOSTIC GRAPH 2: RESIDUALS VS PREDICTED VALUES


ods graphics / imagename="02_residuals_vs_predicted_values";

title1 "Residuals vs Predicted Values";
title2 "Checking Linearity and Constant Variance";

proc sgplot data=diag_soldprice noautolegend;

    scatter x=Predicted_Value y=Residual /
        markerattrs=(symbol=circlefilled size=9 color=CX0891B2)
        transparency=0.15;

    loess x=Predicted_Value y=Residual /
        smooth=0.50
        lineattrs=(color=CX111827 thickness=3);

    refline 0 / axis=y
        lineattrs=(color=CXDC2626 thickness=2 pattern=shortdash)
        label="Zero residual line"
        labelattrs=(color=CXDC2626 size=9 weight=bold);
```

```
    xaxis label="Predicted Sale Price (  )"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);

    yaxis label="Residual"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);

    format Predicted_Value pound12.0;
run;


  25. DIAGNOSTIC GRAPH 3: RESIDUAL DISTRIBUTION


ods graphics / imagename="03_residual_distribution";

title1 "Distribution of Residuals";
title2 "Checking Approximate Normality";

proc sgplot data=diag_soldprice noautolegend;

    histogram Residual /
        fillattrs=(color=CXDBEAFE)
        outlineattrs=(color=CX1E40AF)
        transparency=0.05
        nbins=18;

    density Residual /
        type=normal
        lineattrs=(color=CX111827 thickness=3);

    refline 0 / axis=x
        lineattrs=(color=CXDC2626 thickness=2 pattern=shortdash)
        label="Zero"
        labelattrs=(color=CXDC2626 size=9 weight=bold);

    xaxis label="Residual"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);

    yaxis label="Frequency"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);
run;


  26. DIAGNOSTIC GRAPH 4: STUDENTIZED RESIDUALS VS LEVERAGE


ods graphics / imagename="04_studentized_residuals_vs_leverage";

title1 "Studentized Residuals vs Leverage";
title2 "Checking Unusual and High-Leverage Observations";

proc sgplot data=diag_soldprice noautolegend;

    bubble x=Leverage y=Student_Residual size=Cooks_D /
```

```
        fillattrs=(color=CX7C3AED transparency=0.35)
        lineattrs=(color=CX4C1D95)
        bradiusmin=4
        bradiusmax=14;

    refline 0 / axis=y
        lineattrs=(color=CX6B7280 thickness=1);

    refline -2 2 / axis=y
        lineattrs=(color=CXDC2626 thickness=2 pattern=shortdash)
        label=("-2" "+2")
        labelattrs=(color=CXDC2626 size=9 weight=bold);

    xaxis label="Leverage"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);

    yaxis label="Studentized Residual"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);
run;


  27. DIAGNOSTIC GRAPH 5: COOK'S DISTANCE


ods graphics / imagename="05_cooks_distance";

title1 "Cook's Distance by Observation";
title2 "Checking Influential Observations";

proc sgplot data=diag_soldprice noautolegend;

    needle x=Observation y=Cooks_D /
        lineattrs=(color=CX2563EB thickness=2);

    refline 0.02 / axis=y
        lineattrs=(color=CXDC2626 thickness=2 pattern=shortdash)
        label="Reference line: 0.02"
        labelattrs=(color=CXDC2626 size=9 weight=bold);

    xaxis label="Observation Number"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);

    yaxis label="Cook's Distance"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);
run;

  28. DIAGNOSTIC GRAPH 6: STUDENTIZED RESIDUALS BY OBSERVATION


ods graphics / imagename="06_studentized_residuals_by_observation";

title1 "Studentized Residuals by Observation";
title2 "Checking Outlying Residuals Across the Dataset";

proc sgplot data=diag_soldprice noautolegend;
```

```

    scatter x=Observation y=Student_Residual /
        markerattrs=(symbol=circlefilled size=8 color=CX059669)
        transparency=0.15;

    series x=Observation y=Student_Residual /
        lineattrs=(color=CX059669 thickness=1)
        transparency=0.45;

    refline 0 / axis=y
        lineattrs=(color=CX6B7280 thickness=1);

    refline -2 2 / axis=y
        lineattrs=(color=CXDC2626 thickness=2 pattern=shortdash)
        label=("-2" "+2")
        labelattrs=(color=CXDC2626 size=9 weight=bold);

    xaxis label="Observation Number"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);

    yaxis label="Studentized Residual"
        grid
        gridattrs=(color=CXE5E7EB)
        valueattrs=(size=9)
        labelattrs=(size=11 weight=bold);
run;


  29. TOP 10 INFLUENTIAL OBSERVATIONS


proc sort data=diag_soldprice out=top_influential;
    by descending Cooks_D;
run;

title1 "Top 10 Influential Observations";
title2 "Ranked by Cook's Distance";

proc print data=top_influential(obs=10) label noobs;

    var Observation
        SoldPrice
        Predicted_Value
        Residual
        Student_Residual
        Leverage
        Cooks_D;

    format SoldPrice Predicted_Value Residual pound12.0
           Student_Residual 8.3
           Leverage         8.4
           Cooks_D          8.4;
run;


  30. SAVE FINAL CLEANED DATASET


data MM711.housing_clean;
    set housing_clean;
run;

title1 "Final Cleaned Dataset Saved in MM711 Library";

```

```
proc contents data=MM711.housing_clean varnum;
run;


  31. CLOSE OUTPUT


title;
footnote;

ods html close;
ods listing close;
ods graphics off;
```